# Air New Zealand
# Flight Booking System

*Data Process & Software Modelling . Project Phase 1 – Presentation*

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# Organization Background

Air New Zealand is one of the biggest aviation industry in the Asia-Pacific and also the national airline in New Zealand

Employs 11,700 people globally.

It provides both domestic and international flight services.

Handles large volume of daily operations, including flight scheduling, passenger's booking, and customer service management.

The airline heavily relies on digital systems to manage flight, bookings, and customer interactions.

# Current System

- Air New Zealand currently relies on:
  - fragmented tools
  - manual processes

# Challenges & Inefficiencies

Disconnected Systems

Manual Processes

No real-time updates

Data inconsistency

Poor coordination

Weak customer experience

# Flight Booking System

- To addressed these challenges, there is a clear need for a centralized, integrated flight booking system that can:
  - Provides a single platform for managing flights, bookings, and customer data.
  - Automate key processes such as booking, cancellations, and seat management
  - Enable real-time updates and accurate tracking of flight and booking information
  - Improve coordination between staff and departments
  - Enhance the overall customer experience through faster and more reliable services.

# System Design

- Layered Architecture
- Role-based access
- Centralized system
- OOP-based design

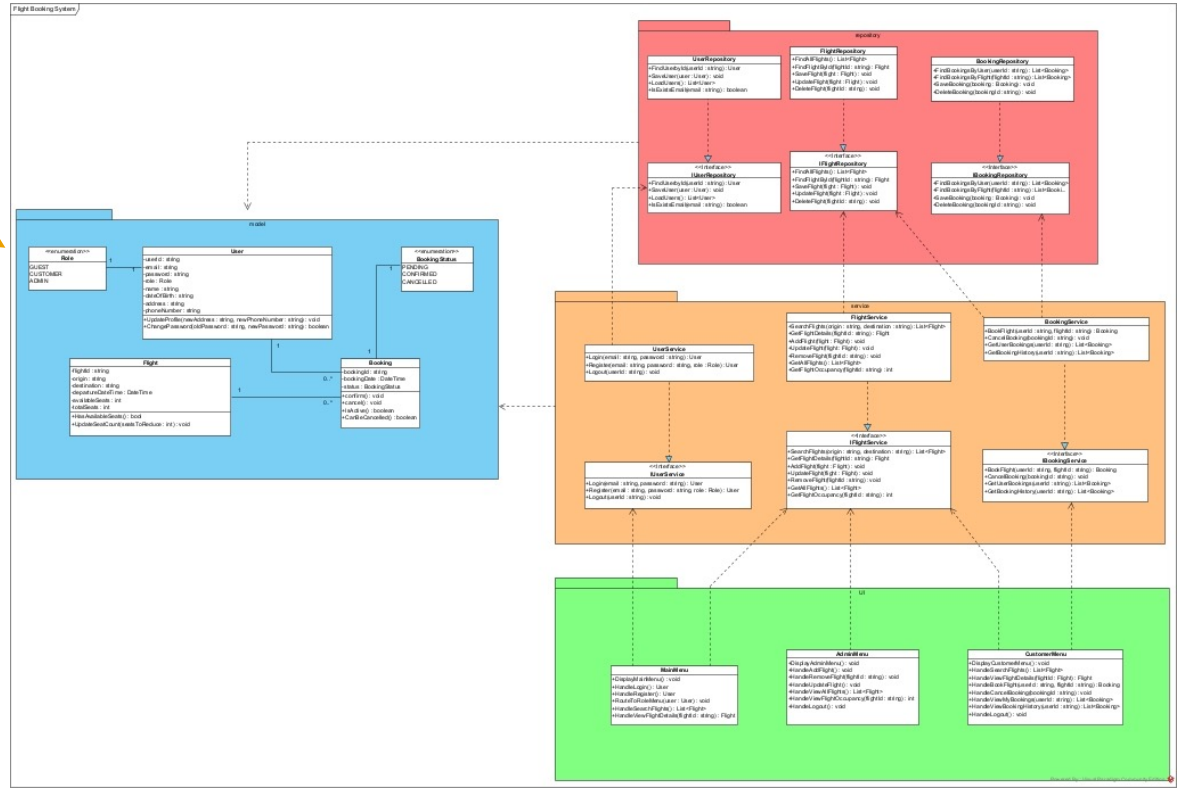
# UML Class Diagram

Repository Layer

Model Layer

Service Layer

UI Layer



# Live Demo

- **Core Functionalities:**
  - User authentication
    - Login
    - Logout
    - Register
  - Role-base Access
  - Flight management
    - Adding Flight
    - View all flights
  - And booking operations
    - Search flights
    - View flight details
    - Book flights
    - View bookings

# Next Phase

- Fully functional flight booking system implementing all business rules
- Add some users such as (Front Desk Staff, Loyal Customer, Flight Manager)
- Ticket Generator
- Payment Processing Simulation